

Reinforcement Learning for Museum Visitor Flow Crowd-Level Interventions and the Optimal Visit at the Uffizi Gallery

Marco Canal

Reinforcement Learning, Barcelona School of Economics

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Why Uffizi Crowding Is a Real Problem

The Uffizi takes roughly *5,000 visitors per day* across *98 rooms*. Demand collapses onto four masterpieces: Botticelli's *Spring* and *Venus* (A11/A12), *Leonardo* (A35), *Raphael / Michelangelo* (A38). The rest of the collection sits nearly empty.

Two problems live in one museum:

- a *population game* (the management problem): how to redistribute demand;
- a *sequential decision* (the visitor problem): when and how to see the masterpieces under crowd uncertainty.

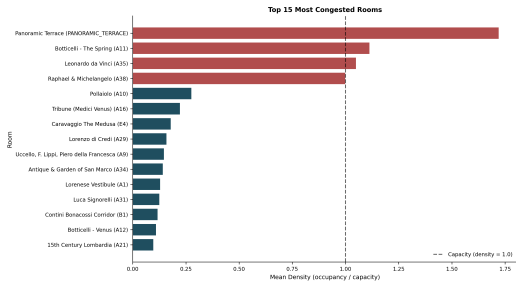
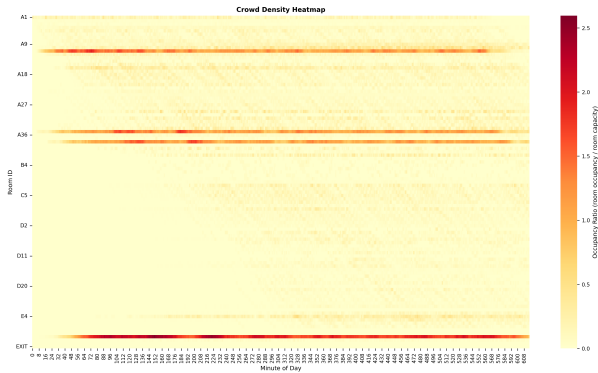
The crowd is not monolithic:

Segment	Share
Instagram tourist (selfie run)	60%
Normal tourist	30%
Art lover (connoisseur)	10%

Different tastes, dwell times, and crowd-aversion, all sharing the same four bottlenecks.

The Status Quo: Masterpieces Overcrowded, the Rest Empty

A single simulated day, no interventions. Left: a time-by-room density heatmap. Right: the most-congested rooms.



The masterpiece rooms run far above capacity while the B, C, and D wings stay quiet. This *imbalance*, not a global shortage of space, is the target.

Two Streams, One Project

The project is deliberately split into two *separate* streams that meet only at the end.

Stream 1: crowd simulator (museum-wide).

- Runs the whole 60/30/10 population for a day.
- Compares the museum *as-is* vs under an *11-lever* intervention bundle.
- Produces *welfare* and *revenue*.

Stream 2: RL on top (the optimal visit).

- Takes the intervened world as given.
- Trains one agent per profile to make the *optimal booking*.
- Reads off the policy: *book early, earlier when busier*.

The gate between them: a director will only adopt interventions that do **not** collapse revenue. *Pass the gate first, then study the optimum.*

The Decision Problems as MDPs

Everything is framed as a Markov decision process $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$.

	Foundations / navigation	RAMA booking (centerpiece)
State s	room, time, local crowd, visited-mask	booking phase, lead options, slot fill, secured windows, position
Action a	stay / move to a neighbour	lead time $\in \{1, 7, 21, 35\}d$, which masterpieces, then pace
Reward r	importance $\times$ crowd-discount – time	appreciation banked at check-in inside the booked window
Discount	$\gamma = 0.99$ (toy), 0.995 (deep)	$\gamma = 0.999$ (reward lands days later)

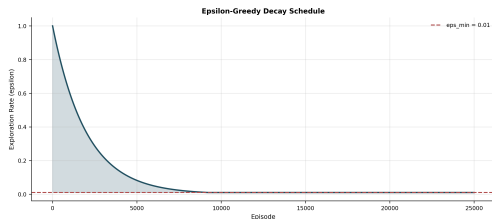
Why RL and not plain optimisation? The crowd is non-stationary and only partially observed; the payoff is delayed; the visitor must trade present dwell against future congestion. That is a sequential decision under uncertainty.

We Start Small: a 12-Room Toy Museum

Before the full museum, we prove the core idea on a *12-node toy graph* where the state space is small enough for an exact table.

- Nodes: ENTRY, two corridors, the masterpieces (A11/A12, A16, A35, A38, E4), two minor rooms, EXIT.
- Crowds are *deterministic Gaussian waves*: Botticelli peaks at 35% of the day, the Tribune at 40%, Leonardo at 50%, Raphael at 55%.

The hypothesis: the best policy is **temporal arbitrage**, visit a masterpiece in the valley before or after its crowd peak, not when the guidebook sends everyone.



Exploration schedule: ϵ decays $1.0 \rightarrow 0.01$ multiplicatively ($\times 0.9995/\text{episode}$), annealing from random to greedy.

Q-Learning, in Detail

A model-free, off-policy, value-based method. We learn $Q(s, a)$, the expected return of taking a in s then acting greedily, by bootstrapping toward the Bellman target.

The one-step TD(0) update

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[\underbrace{r + \gamma \max_{a' \in \mathcal{A}(s')} Q(s', a')}_{\text{TD target}} - Q(s, a) \right]$$

State (a 5-tuple, $\approx 123k$ cells):

room $\times$ time-bin(10) $\times$ Bott-crowd(4) $\times$ Leo-crowd(4) $\times$ visited-mask(64).

Reward encodes the art lover:

$$r = \frac{\text{importance} \cdot \text{novelty}}{1 + \alpha_A \text{density}^2} - \text{step cost}$$

novelty halves each extra minute (satiation); $\alpha_A=6$.

Hyperparameters

- learning rate $\alpha = 0.1$
- discount $\gamma = 0.99$ (far-sighted: late rooms still matter)
- ϵ -greedy, masked to valid moves only
- max in the target is also masked, so the agent never bootstraps off illegal moves

Double Q-Learning, in Detail

Vanilla Q-learning *overestimates*: the same noisy Q both *picks* and *scores* the next action, and by Jensen's inequality $\mathbb{E}[\max_a Q] \geq \max_a \mathbb{E}[Q]$. Double Q-learning (van Hasselt, 2010) decouples the two.

Two tables, a coin flip decorrelates selection from evaluation

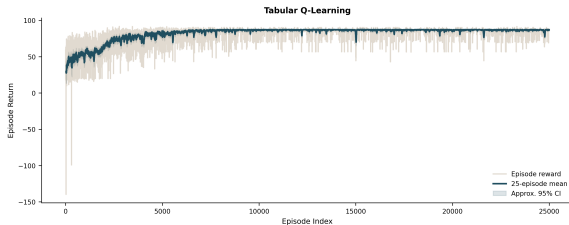
With probability $\frac{1}{2}$ update Q_A , else Q_B :

$$Q_A(s, a) \leftarrow Q_A(s, a) + \alpha \left[r + \gamma Q_B(s', \arg \max_{a'} Q_A(s', a')) - Q_A(s, a) \right]$$

Q_A chooses the action; the *other* table Q_B scores it. At decision time we act on the average $\frac{1}{2}(Q_A + Q_B)$.

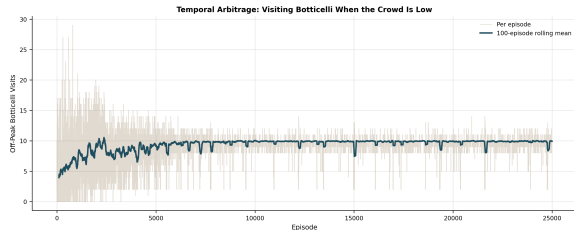
Honest result (next slides): on this **deterministic** toy there is no maximisation bias to remove, so across 5 seeds vanilla and Double-Q come out *statistically tied* (83.3 vs 83.1, overlapping bands). The variant earns its keep under noise, not here. Reporting both, with error bars, is the point.

Foundations Result I: Temporal Arbitrage Is Learned



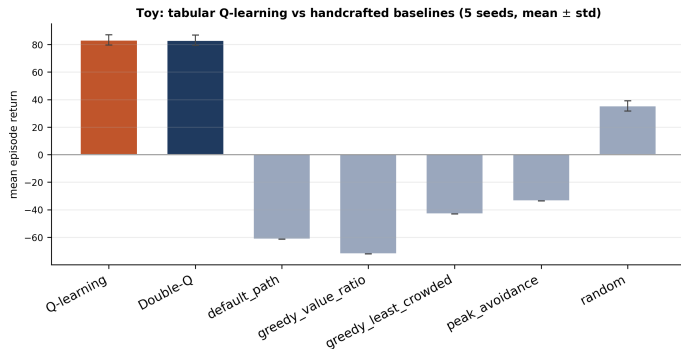
Episode return climbs from ~ 40 to ~ 85 and converges. The band is the 25-episode rolling 95% CI.

The agent is never told the crowd schedule. It discovers, by trial and error, that *when* you visit a masterpiece matters as much as *whether* you visit it.



Off-peak Botticelli visits rise over training: the agent learns to enter A11/A12 in the crowd valley, from samples alone.

Foundations Result II: Beating Every Handcrafted Baseline



- Q-learning (83.3 ± 3.8) beats every hand rule by a wide margin: the best, random, reaches only ~ 35 ; the structured rules go negative.
- Double-Q (83.1 ± 3.7) is *statistically tied* with vanilla: no maximisation bias to remove on a deterministic graph, so it neither helps nor hurts.
- *Takeaway*: the learned timing dominates every hand rule across all 5 seeds.

Why the Table Breaks on the Real Museum

The toy table has $\sim 123\text{k}$ cells. The real museum does not fit:

- *98 rooms*, ~ 100 actions per step, a continuous crowd density per room, and a ~ 400 – 500 -dimensional observation.
- A tabular state space would be astronomically large and almost entirely unvisited, so no generalisation across similar states.

The fix is function approximation: replace the table $Q(s, a)$ with a neural network $\pi_{\theta}(a|s)$ and $V_{\theta}(s)$, trained by policy gradient. We use **MaskablePPO**, with vanilla PPO and DQN as controls.

MaskablePPO, in Detail (1/2): the Objective

PPO is an *on-policy actor-critic*. The actor π_θ proposes moves; the critic V_θ scores states. It improves the policy with a *clipped* surrogate that forbids destructively large updates.

Clipped surrogate objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad \rho_t = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$$

Advantages use *GAE*: $\hat{A}_t = \sum_l (\gamma \lambda)^l \delta_{t+l}$, $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$.

- clip range $\epsilon = 0.2$ keeps the policy ratio in $[0.8, 1.2]$;
- $\gamma = 0.995$ (long, 615-minute episodes), $\lambda = 0.98$ (low-bias advantages);
- network: `MlpPolicy`, two hidden layers of 256; entropy bonus 0.005.

MaskablePPO, in Detail (2/2): Action Masking

The museum is a *graph*: from any room only a handful of moves are legal, but the action head has ~ 100 outputs. Without help, $\sim 80\%$ of actions are illegal at every step and the policy must *learn* that from penalties, wasting gradient.

Masking = restrict the action set before the softmax

For invalid actions $a \notin \mathcal{A}(s)$, set the logit $z_a \rightarrow -\infty$, so

$$\pi_{\theta}(a | s) = \frac{\exp z_a \mathbb{I}[a \in \mathcal{A}(s)]}{\sum_{a' \in \mathcal{A}(s)} \exp z_{a'}} = 0 \text{ for invalid } a.$$

- Mathematically equivalent to a state-dependent action space, with no probability mass ever spent on illegal moves.
- Implemented with sb3-contrib's `ActionMasker`; the same mask is passed at training and evaluation.
- *This is the single most important architectural choice*, quantified in the ablation later.

The Curriculum and the DQN Control

Three-stage crowd curriculum.

Training directly at 5,000 visitors is too noisy early on, so we ramp the crowd:

- Stage 1 (15% of steps): $0.36\times$ peak crowd
- Stage 2 (35%): $0.64\times$ peak
- Stage 3 (50%): full peak crowd

Episodes are always the *full open day* (615 min); only the crowd grows. `random_start` during training (exploring starts) escapes the entrance-room trap.

DQN, the value-based control.

Off-policy, learns Q by minimising the Bellman error against a *target network*:

$$\left(r + \gamma \max_{a'} Q_{\theta^-}(s', a') - Q_{\theta}(s, a)\right)^2$$

- replay buffer 200k, batch 128, $\gamma = 0.995$;
- no masking, no curriculum (a fair, reasonable default).

Including a value-based method shows the masking benefit is not PPO-specific.

Evaluation Protocol: Common Random Numbers

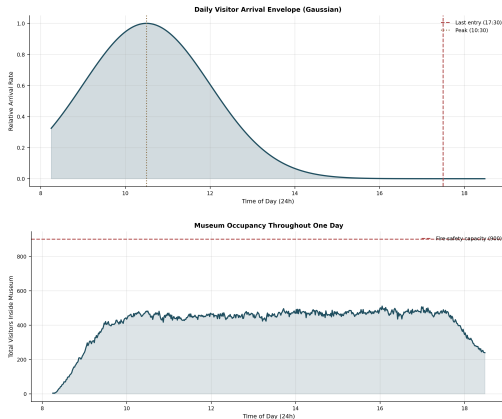
Fair comparison demands that every model is scored on the *exact same* crowds.

- *CRN eval*: episode i always runs on crowd scenario seed $900000+i$, identical regardless of training seed. The reward gap then reflects the *policy*, not which random crowds it happened to face.
- 50 evaluation episodes; we report mean and standard error.
- *We score the stochastic policy* (sample, not argmax). Under the egress reward the greedy argmax collapses into a non-exiting loop; the sampled policy exits reliably. Scoring the policy as trained is the honest choice (diagnosed later).
- *Apples-to-apples*: every algorithm gets the same step budget (up to 1M) and the same CRN eval. “10k vs 1M would be unfair.”

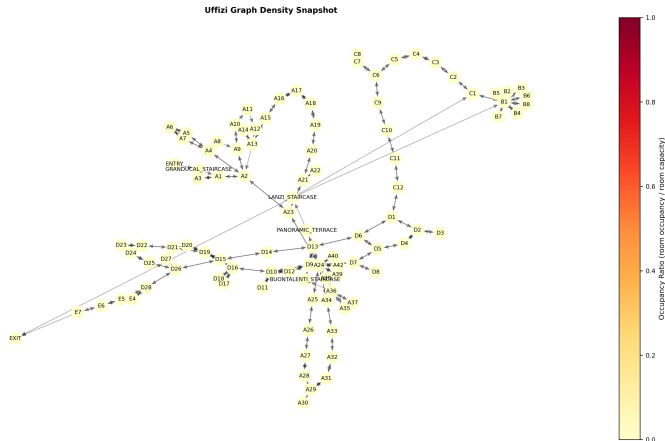
An Agent-Based Simulator of the Whole Museum

- Minute-step, 98 rooms, capacities scaled from floor-plan area (*never altered*).
- Visitors arrive on a *Gaussian schedule*, book or walk in, and move by taste under crowd pressure.
- The 60/30/10 segments differ in dwell time, masterpiece pull, and crowd-aversion.
- Output per day: a density tensor plus *welfare*, *revenue*, and *experience* metrics.

This is the world. Both the interventions and (later) the RL agent run on top of it.



Where the Crowds Form



The museum graph coloured by mean density. The hot nodes are exactly the four masterpiece rooms and the corridors feeding them; the periphery is cold. The intervention design targets this gradient, not the total headcount.

Eleven Interventions: Pigovian, Not Restrictive

Design principle (Ostrom / Williamson): *price and reshape the externality*, do not lock people out of rooms.
The full bundle:

- *RAMA*: reservation-anchored masterpiece access
- *Secondary-attractor enrichment* (light up minor rooms)
- *Extended hours* (7:00–19:00)
- *Dynamic pricing* (peak surcharge, off-peak discount)
- *Resident annual pass*
- *Tour-group cap* (10)
- *Timed entry*
- *Quiet hours*
- *Per-person group surcharge*

None restrict access; all redistribute demand or price congestion.

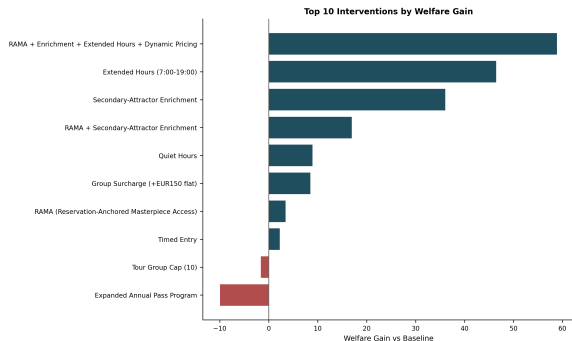
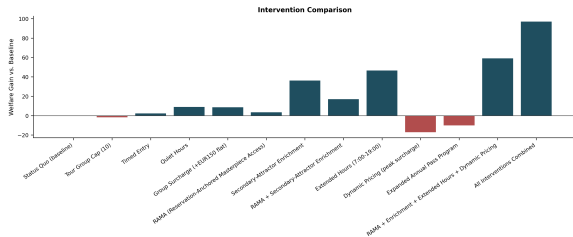
RAMA: Reservation-Anchored Masterpiece Access

The keystone lever, and (later) the object the RL agent learns to use.

- The three floor-2 masterpieces (Botticelli, Leonardo, Raphael) are gated by *booked 10-minute time windows* with a per-room capacity ledger (A11=55, A35=52, A38=53). Caravaggio stays free-flow.
- You book a general-entry ticket plus the masterpiece slots you want, in advance.
- No walk-ins to the masterpieces: if a slot is gone, it is gone, so popular slots on busy days sell out at long lead times.
- Effect on the crowd: the masterpiece rooms run *at*, not above, capacity, and the queueing chaos disappears.

RAMA is what turns “when to visit” into “when to book.”

Screening the Interventions

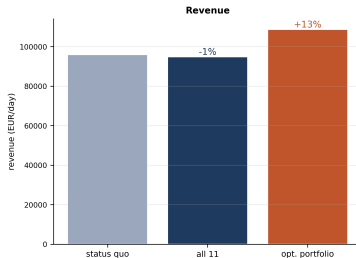
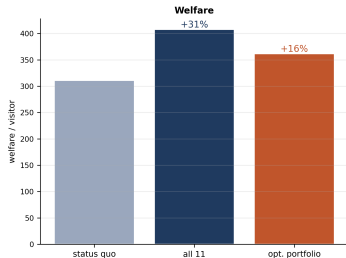


Each lever is evaluated alone against the status quo. RAMA and enrichment carry most of the welfare gain; the pricing levers carry the revenue. The portfolio optimiser then searches combinations.

The Director Gate: Welfare Up, Revenue Preserved

A director will only adopt interventions that do *not* collapse revenue. So before any RL, we run the whole population, as-is vs all eleven, and check.

Museum-wide director gate (as-is crowd): all 11 lift welfare with revenue preserved; the optimized subset raises both

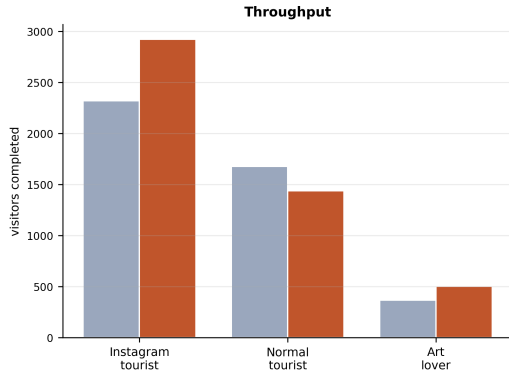
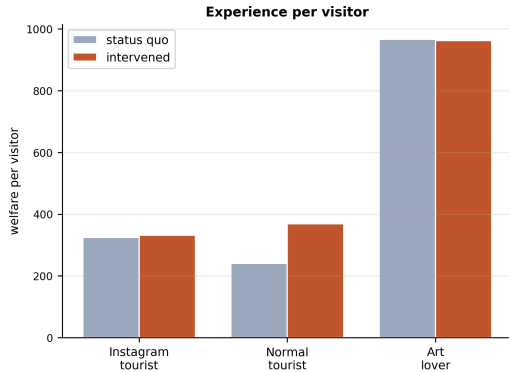


- All 11: welfare **+31%**, revenue *preserved* (−1%, no collapse).
- Optimized subset: welfare **+16%** *and* revenue **+13%**, both up.
- Peak Botticelli density **1.82** → **1.00**.

The gate passes: a large welfare gain at no revenue cost. *Green light to study the optimal visit.*

Who Gains? Every Segment

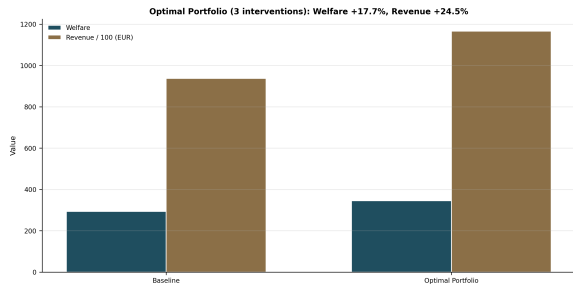
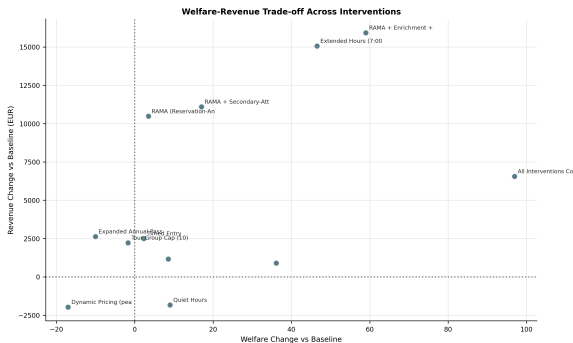
Museum-wide by segment (as-is crowd): normal tourists gain most per visit; more Instagram tourists and art lovers get through



Normal

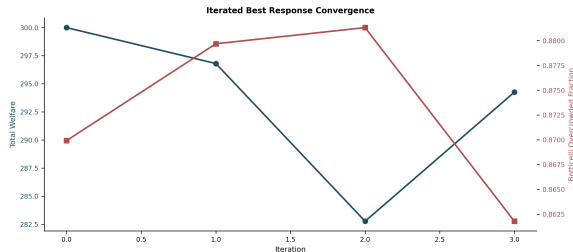
tourists gain most per visit (+49% to +58%); *Instagram tourists* and *art lovers* gain through *throughput*, more of them complete a visit once the masterpieces are decongested. No segment is made worse off.

The Win-Win Frontier and the Portfolio

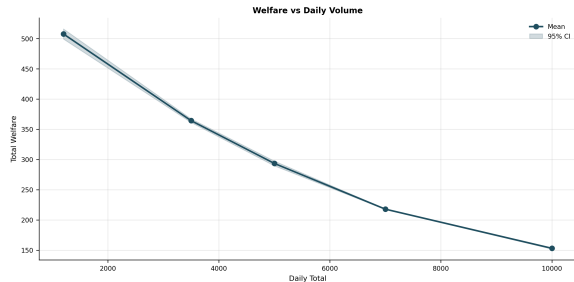


The optimised portfolio drops the heavy demand-suppressors and keeps RAMA, enrichment, extended hours, dynamic pricing, the group cap, and the resident pass: it lands in the *upper-right*, welfare **up** and revenue **up** together.

Population Game: Equilibrium and Sensitivity



Iterated best response converges; the gap to the social optimum is the Price of Anarchy the interventions close.



Welfare vs daily volume: the gain is robust across crowd sizes, widest exactly when the museum is busiest.

From the Average to the Optimum

The museum-wide result is an *average* over the whole population. With the interventions cleared as director-safe, the RL question is the sharper one:

*Not the average welfare, but the **top** welfare: what does an optimal art lover and an optimal normal tourist **do** in the intervened museum?*

We train one RL agent per profile, on top of the same crowd simulator, at three crowd levels (500 / 2,500 / max), and read off its policy.

The Decision Is the Booking, Not the Navigation

A real visitor has a map and is guided by staff. Navigation is not the hard problem, and pretending it is would be a fake RL task. The genuine learned decision is the *RAMA booking*:

- *Lead time*: how many days ahead to book (1, 7, 21, 35).
- *Which masterpieces* to reserve.
- then *pace the visit* to arrive inside each booked window.

Setup:

- MaskablePPO, $\gamma = 0.999$ (the booking reward lands at check-in, days later).
- Two profiles: art lover, normal tourist.
- Three crowds: 500 / 2,500 / max.
- A *fill curve*: popular slots need a longer lead on busier days.

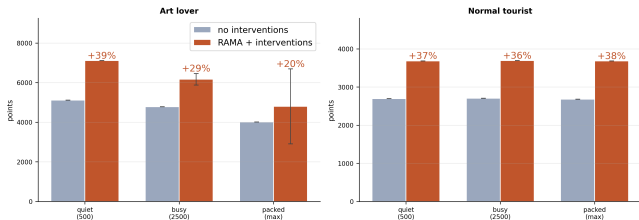
The Booking MDP

A two-phase episode. *Phase 1 (booking)*: choose a lead time, then accept or skip each masterpiece slot, resolved against the crowd-dependent fill curve. *Phase 2 (the visit)*: walk the museum and check in at each booked window.

- *Reward structure*: a masterpiece pays *nothing* until you check in inside its window; minor rooms pay their (small, nonzero) encoded value; a kept appointment pays a check-in bonus.
- *The fill curve* ties lead time to crowd: a slot is available iff $\text{lead} \geq \text{lead}_{\max} \cdot \text{popularity} \cdot (\text{daily}/2500)$.
Busier day $\Rightarrow$ longer lead required.
- *Why book-early is non-trivial*: booking late at a packed museum means the good slots are sold out, so you miss masterpieces entirely. Booking early secures well-spaced slots. There is no downside to early; the agent must *discover* that.

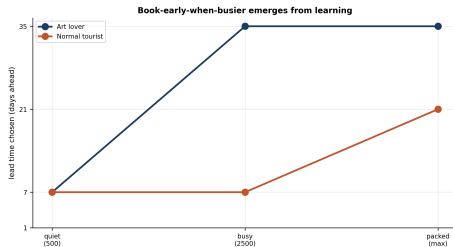
Result: Book Early When Busier, and It Pays

RAMA booking: intervened vs matched baseline (5 seeds, mean $\pm$ std; 3/3 masterpieces secured)



Intervened (RAMA agent) vs matched no-intervention baseline; 3/3 masterpieces secured at every crowd.

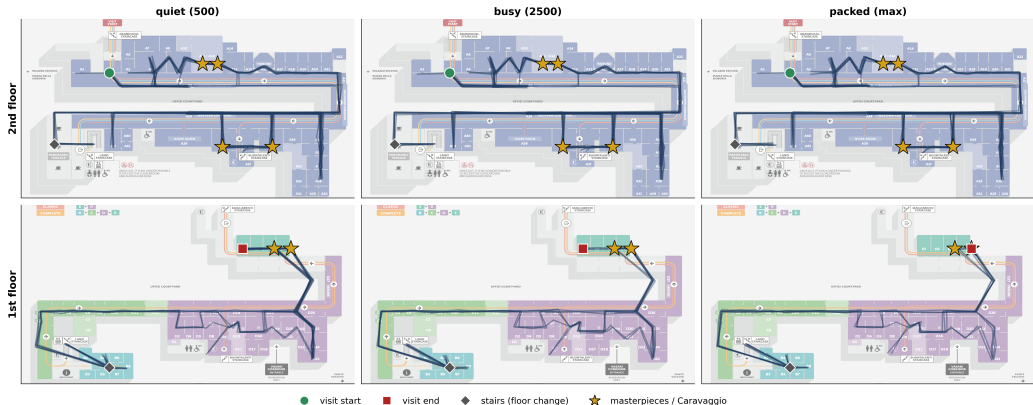
- Both profiles beat baseline (means over 5 seeds): +20% to +39% (art), +36% to +38% (tourist).
- Lead jumps 7 $\rightarrow$ 35 days as the day fills; booking *late* at a packed museum is worse than not booking.



Lead time rises with the crowd: book-early emerges from learning.

The Learned Route on the Real Floor Plan

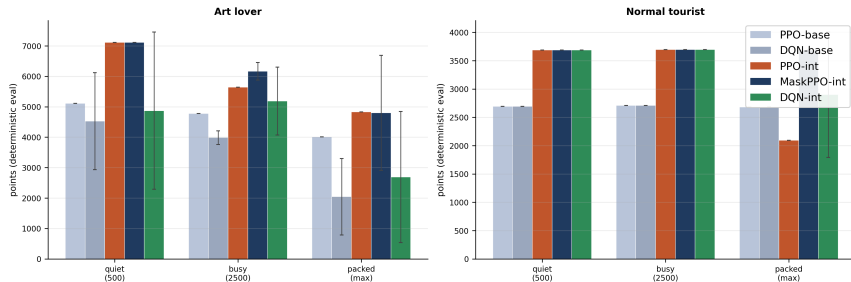
Art lover walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



Art lover, intervened, eight rollouts per panel, across the three crowds. One continuous visit: enters at A1, sweeps the second-floor masterpieces (gold stars), descends to the first floor for Caravaggio, exits. Every move is a one-hop neighbour (no teleports); all three masterpieces are hit inside their booked windows.

Which Algorithm Wins, and Why Masking Matters

Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean $\pm$ std)



- *MaskablePPO* is the clear winner: stable across seeds, finds book-early, best or tied-best at every crowd.
- *Masking is decisive where book-early matters*: at the packed tourist, unmasked PPO collapses to **2093**, below the no-booking baseline (2680), while MaskablePPO holds **3685**, on every seed.
- *DQN* is crowd-fragile and high-variance on the art lover: its baseline falls 4527 $\rightarrow$ 2044 as the crowd grows, with seed std up to $\pm 2,600$ (the wide error bars).

How to Read This Section

The project was months of being wrong in instructive ways. Each of the following slides follows the same anatomy:

- S** *Symptom* : what looked broken.
- D** *Diagnosis* : the actual cause, often not where I first looked.
- F** *Fix* : the change that worked.
- L** *Lesson* : what it taught about RL or about modelling.

These failures, and the diagnoses, are the project. The final numbers are just where the failures stopped.

Challenge 1: “The Guidebook Tourist Does Not Exist”

- S** *Symptom.* Early visitor model assumed a single rational “guidebook” tourist who tours the collection methodically. The simulated crowd looked nothing like the real Uffizi.
- D** *Diagnosis.* Wrong population. The modal visitor is an *Instagram tourist*: four famous rooms, a terrace selfie, gone in 30–60 minutes. Dwell at a masterpiece is bottlenecked by selfies and crowd-navigation, not appreciation.
- F** *Fix.* Rebuilt the population as *60% Instagram / 30% normal / 10% art lover*, each with distinct dwell, masterpiece pull, and crowd-aversion.
- L** *Lesson.* Get the agents right before the mechanism. A beautiful optimiser over the wrong population optimises nothing.

Challenge 2: A Welfare Metric That Punished the Wrong People

- S** *Symptom.* Under RAMA, total welfare came out *below* the no-RAMA baseline (208 vs 244), which made the keystone intervention look harmful.
- D** *Diagnosis.* The objective double-counted: a visitor who knew *at booking time* that a masterpiece was full was still penalised for “missing” it, as if surprised on the day.
- F** *Fix.* Redistribute the lost “wantedness mass” onto the rooms the visitor *can* see; score welfare as visitor-integrated importance under a quadratic crowd discount, per attempted visitor.
- L** *Lesson.* The objective must match what the agent actually knows and chooses. A metric that penalises foreseen, accepted trade-offs is measuring the wrong thing.

Challenge 3: The Big Pivot, Navigation Was the Wrong Task

- S** *Symptom.* Months of fighting a navigation agent (exit, dwell, coverage) produced a brittle policy and a nagging worry: is routing-with-a-map even a legitimate RL problem?
- D** *Diagnosis.* It is not the interesting decision. The visitor *has* a map and is guided by staff; optimising the walk is a shortest path dressed up as RL, a trivial state space.
- F** *Fix.* Reframe the task around the *booking*: lead time, which masterpieces, and spacing under a crowd-dependent fill curve. Navigation becomes auto-pacing underneath.
- L** *Lesson.* Choosing the MDP is the modelling. Spend the effort on the decision that is genuinely uncertain and consequential; the right formulation is more than half the project.

Challenge 4: The Exit Bug, and an Unlearnable Reward

- S** *Symptom.* The agent never left the museum: it squatted in a room until the episode truncated. The obvious fix, zero out all reward if you fail to exit, made it *worse*.
- D** *Diagnosis.* An all-or-nothing terminal wipe is *unlearnable*: the return is flat across almost every policy, so the gradient is zero, nothing points toward the door.
- F** *Fix.* Replace the wipe with a *dense* egress signal: a per-step pressure that grows with distance-to-exit as closing nears, plus a finite (not infinite) no-exit penalty and an exit bonus.
- L** *Lesson.* Sparse all-or-nothing rewards give the optimiser nothing to climb. A shaped, dense signal that points the right way is what makes a behaviour learnable.

Challenge 5: The Agent Could Not See What It Needed to Decide

- S** *Symptom.* “Stay until satisfied, then move on” never stabilised: the agent over- or under-dwelt rooms seemingly at random, even with the right reward.
- D** *Diagnosis.* The rule depends on *how much value I have already extracted here*, but that progress was *not in the observation*. The decision was non-Markov in the state the policy could see, an unobservable it was being asked to act on.
- F** *Fix.* Add a per-room *appreciation-progress* vector to the observation (state grows to $5N+3$ dims), so “have I seen enough of this room?” is a function of what the policy observes.
- L** *Lesson.* The state must contain everything the optimal action depends on. A reward that hinges on hidden progress turns the task into a POMDP; expose the variable rather than hoping the network reconstructs it.

Challenge 6: The Fake Art Lover

- S** *Symptom.* The “art lover” speed-ran the masterpieces and camped the calm Caravaggio room; it also valued every minor room at exactly zero.
- D** *Diagnosis.* Two reward-modelling errors. (1) The crowd penalty was so harsh that a crowded Botticelli scored worse than skipping it, so the agent skipped. (2) An importance floor zeroed minor rooms, when rooms carry real encoded magnetism and value.
- F** *Fix.* A *gentle* crowd discount (value = importance $\times \frac{1}{1+\alpha \text{ dens}^2} \times \text{novelty}$, small α): a crowded masterpiece still beats skipping it; minor rooms keep a small positive value.
- L** *Lesson.* Calibrate the reward to the behaviour you are modelling. A connoisseur who skips a crowded Botticelli, or values a minor room at zero, is not a connoisseur.

Challenge 7: Reward Shaping That Backfired

- S** *Symptom.* Well-meant shaping terms made behaviour *worse*: a “boredom” penalty for lingering produced *looping*, and a strong tour-completion pull produced *ping-pong* between two rooms.
- D** *Diagnosis.* Each term opened a loophole the optimiser exploited. Circling was cheaper than exiting under the boredom penalty; an over-weighted tour pull overwhelmed the value signal and oscillated.
- F** *Fix.* Remove the boredom penalty (weight 0) and reduce the tour pull to a gentle nudge (weight 1); let the satiation / novelty term carry “stop lingering.”
- L** *Lesson.* Dense shaping is double-edged: every extra term is a new objective the agent will game. Prefer the *minimal* shaping that points the right way, and stress-test each term for the perverse incentive it creates.

Challenge 8: The Decline-Trap (a Do-Nothing Local Optimum)

- S** *Symptom.* Given a free “decline” button, *both* profiles collapsed to declining all three masterpieces, even against a -1200 immediate regret and after a curriculum that first taught booking pays $\sim +2000$.
- D** *Diagnosis.* A free decline is a classic RL *do-nothing trap*: skipping is always immediately safe, the payoff for booking is delayed, so the optimiser parks on the safe action.
- F** *Fix.* Treat “always books the three masterpieces” as a *type characterisation*, not museum coercion: nobody travels to Florence and skips all of Botticelli, Leonardo, and Raphael. The genuine learned decision is the *lead time*.
- L** *Lesson.* A voluntary action that is always immediately safe but only delayed-rewarding will be abused by the optimiser unless the formulation itself rules it out.

Challenge 9: The Learned-Prior Runaway

- S** *Symptom.* The agent's estimate of its own arrival time drifted later and later each update, until it "planned" to arrive at closing, even on an efficient hand-coded walk and at a fixed crowd.
- D** *Diagnosis.* A self-referential feedback loop: the arrival prior was an online average of the agent's *own* past arrivals, an alarm clock set to your own wake-ups, so any drift compounded.
- F** *Fix. Freeze the prior:* compute the arrival estimate *once* from reference walks ("brief memory"), then hold it fixed during training ($\alpha_{arr}=0$).
- L** *Lesson.* Beware state that is a function of the policy's own history feeding back into the policy. Break the loop by freezing the reference.

Challenge 10: Book-Late $\rightarrow$ Book-Early (the Decisive Trick)

- S** *Symptom.* Even after everything above, book-early was only *half*-learned: a flat lead-time distribution, 2.5/3 masterpieces at the packed crowd. The agent also idled ~ 120 minutes waiting for windows.
- D** *Diagnosis.* No immediate signal connected an off-pace booking to its eventual cost; the credit for booking well was spread days into the future, where the gradient could not find it.
- F** *Fix.* Add an *immediate* off-pace penalty $\text{mismatch}_k \cdot |\text{resolved window} - \text{estimate}|$ at booking time, a local proxy for the distant cost (plus a slot-sized min gap and an early-grace band to remove the manufactured waiting).
- L** *Lesson.* Long-horizon credit assignment is hard; a small immediate penalty that correlates with the distant payoff is what made the lesson learnable: book-35 at the packed museum, 3/3, zero waiting.

Challenge 11: Evaluate the Policy You Actually Trained

- S** *Symptom.* The deterministic (argmax) policy looked degenerate, a camp or a ping-pong loop, even when training reward was healthy, suggesting the agent had learned nothing.
- D** *Diagnosis.* Under the egress shaping the greedy argmax never quite selects EXIT and falls into a non-exiting loop; the *sampled* policy, the one PPO actually optimised, exits reliably. Greedy eval was measuring a policy we never trained.
- F** *Fix.* Score the *stochastic* policy for the navigation agent (as trained); score the clean booking policies *deterministically*, where the argmax books cleanly.
- L** *Lesson.* The evaluation must match the object you optimised. “Deterministic eval” is a convention, not a law; for a policy trained to be stochastic, the argmax can misrepresent what was learned.

Scoreboard: Conceptual Choices That Worked Best vs Worst

Worked best

- *Action masking*: restrict the action set; the single biggest lever.
- *Reframing to booking, not navigation*: the right, non-trivial MDP.
- *Exposing appreciation progress in the state*: made the dwell rule Markov.
- *An immediate proxy for the delayed reward*: solved book-early credit assignment.
- *Dense egress shaping*: replaced an unlearnable terminal wipe.
- *Freezing the self-referential prior*: killed the runaway feedback loop.

Worked worst

- *All-or-nothing exit wipe*: zero gradient, unlearnable.
- *Free decline button*: a do-nothing local optimum.
- *Online self-referential prior*: drifted to closing time.
- *Harsh crowd penalty*: produced a fake, masterpiece-skipping art lover.
- *Boredom penalty*: created a perverse looping incentive.
- *Greedy eval* of a stochastic policy: misrepresented it.

Two Streams, One Story

- 1 *Crowd simulator (museum-wide)*. The eleven-lever Pigovian bundle raises welfare **+31%** with *revenue preserved*; the optimised subset raises both (**+16%** welfare, **+13%** revenue). The interventions are director-safe, no revenue collapse.
- 2 *RL on top (the optimal visit)*. Given that intervened world, the optimal art-lover and normal-tourist agents *book early, and earlier when busier*, secure all three masterpieces, and beat the no-intervention baseline by **+20% to +39%** (art) and **+36% to +38%** (tourist), averaged over 5 seeds.

The average visitor is better off and the museum keeps its revenue; the optimal visitor shows exactly how to use the redesigned museum.

Limitations and What Is Next

- *Five-seed algorithm matrix.* The 30-cell comparison is now averaged over 5 training seeds (mean $\pm$ std). This confirms DQN's high variance on the art lover, and exposes that the art-lover / max-crowd MaskablePPO gain is itself high-variance: 4 of 5 seeds reach $\sim +41\%$, one collapses, so that cell averages $+20\%$ with a wide band.
- *"Always books" is a type assumption,* defensible for these profiles but a modelling choice, not a learned outcome. A richer model would let lead time and participation co-vary.
- *The fill curve is a calibrated model,* not booking-system data; the qualitative lesson (book earlier when busier) is robust to its exact shape.
- *Next: the Vatican Museums.* Same two-stream method, a larger topology, and real booking traces to replace the fill-curve assumption.

Thank you.

Questions?

Code, data, and reproducible figures at
`github.com/mocl20/Uffizi`